Qiskit Fall Fest 2026 — Quantum Chemistry Track

Phase 2 Technical Report: PS-C1 — H■ Ground-State Energy

Team: Straw Hats

Problem Statement: PS-C1

Molecule: H■

Bond length: 0.735 Å

Basis set: STO-3G

Reference FCI energy: −1.1373060357534004 Hartree

1. Objective

The objective of this project is to estimate the ground-state energy of the hydrogen molecule (H■) using quantum-computing techniques and evaluate the effects of qubit mapping, circuit depth, optimization algorithms, finite-shot measurements, and processor connectivity.

The Phase 2 work includes:

• Constructing and verifying the molecular Hamiltonian.

• Comparing Jordan–Wigner and parity mappings.

• Reducing the problem from four qubits to two qubits.

• Performing a 12-run VQE optimization sweep.

• Preparing and verifying Bell states on Processor D.

• Studying finite-shot energy estimation under ideal and noisy conditions.

• Comparing circuit resources on Processors A, B, and D.

The experiments use the organizer-provided chemistry integrals and the supplied starter implementation.

2. Hamiltonian Construction and Qubit Mapping

2.1 Molecular system

The investigated molecule is H■ at an internuclear distance of 0.735 Å using the STO-3G basis set. The system contains two electrons and two spatial orbitals.

The one-electron and two-electron integrals are loaded from the supplied file:

```
data/c1/h2_0.735_integrals.npz
```

The electronic Hamiltonian is constructed using Qiskit Nature. The core-energy constant is included exactly once when calculating the total molecular energy.

The reference full configuration interaction (FCI) energy is:

$E_{FCI} = -1.1373060357534004$ Ha

2.2 Jordan–Wigner mapping

The Jordan–Wigner transformation maps the fermionic Hamiltonian to a four-qubit Pauli operator.

The resulting operator contains 14 non-identity Pauli terms and requires five qubit-wise commuting (QWC) measurement groups.

2.3 Parity mapping and two-qubit reduction

The parity transformation uses the known particle numbers, one alpha electron and one beta electron, to perform two-qubit reduction.

The resulting representation contains two qubits, four non-identity Pauli terms, and two QWC measurement groups.

Mapping	Qubits	Non-identity terms	QWC groups	Exact energy (Ha)	CX on D
Jordan–Wigner	4	14	5	–1.1373060357534004	6
Parity reduced	2	4	2	–1.1373060357533990	2

The parity-reduced energy agrees with the FCI reference to numerical precision.

This demonstrates that the two-qubit reduction preserves the molecular ground-state energy while reducing the number of qubits and measurement groups.

3. VQE Optimization and Convergence

3.1 Ansatz and optimization settings

A hardware-efficient ansatz was used with a parity-mapped two-qubit Hamiltonian and the corresponding Hartree–Fock initial state.

Two ansatz depths were investigated:

• Depth 1: one repetition.

• Depth 2: two repetitions.

Two classical optimizers were compared:

• COBYLA.

• SPSA.

Three random initial parameter seeds were used for each combination. This produced 12 optimization runs.

Initial parameters were sampled from a uniform distribution over the interval $[-\pi, \pi]$.

The exact energy was evaluated using statevector simulation. Transpiled circuit metrics were obtained using the official Processor D configuration.

3.2 Optimization results

Ansatz depth	Optimizer	Seed	Energy error (mHa)	Evaluations	CX on D	Depth on D
1	COBYLA	0	≈0.0000	99	1	8
1	COBYLA	1	≈0.0000	116	1	8
1	COBYLA	2	≈0.0000	106	1	8
1	SPSA	0	1.6399	651	1	8
1	SPSA	1	0.0001	651	1	8
1	SPSA	2	0.0011	651	1	8
2	COBYLA	0	≈0.0000	94	2	13
2	COBYLA	1	≈0.0000	119	2	13
2	COBYLA	2	≈0.0000	143	2	13
2	SPSA	0	0.0004	651	2	13
2	SPSA	1	0.0011	651	2	13
2	SPSA	2	0.0010	651	2	13

The chemical-accuracy threshold is 1.6 mHa.

All six COBYLA runs and five of the six SPSA runs met this threshold.

COBYLA required between 94 and 143 evaluations, whereas SPSA used 651 evaluations in each run. For this small molecular problem, COBYLA achieved the target accuracy with substantially fewer evaluations.

The depth-1 ansatz also required fewer CX gates and had lower transpiled depth than the depth-2 ansatz.

The best depth-1 COBYLA configuration was used for the subsequent shot-based VQE experiment.

4. Bell-State Preparation and Verification

4.1 Objective

The Bell-state experiment evaluates Processor D's ability to prepare and preserve entanglement.

The target state is:

Φ■■ = (	00■ +	11■) / √2

It is prepared using a Hadamard gate on the first qubit followed by a CNOT gate from the first qubit to the second.

Two physical-qubit pairs were investigated:

• Near pair: (0,1), distance 1.

• Far pair: (1,3), distance 2.

Each pair was evaluated under ideal and noisy conditions using 4096 shots per measurement basis.

4.2 Verification method

The experiment measures the XX, YY, and ZZ correlations.

Bell-state fidelity is calculated as:

F = (1 + XX – YY + ZZ) / 4

The fidelity standard error is calculated from the standard errors of the three measured correlations.

4.3 Results

Pair	Distance	Condition	Two-qubit gates	Fidelity	Fidelity SE
(0,1)	1	Ideal	1	1.0000	0.0000
(0,1)	1	Noisy	1	0.9633	0.0021
(1,3)	2	Ideal	4	1.0000	0.0000
(1,3)	2	Noisy	4	0.9557	0.0023

Both pairs achieve ideal fidelity in noiseless simulation.

Under noise, the directly connected pair achieves higher fidelity than the distance-two pair. The far pair requires four two-qubit gates compared with one for the near pair.

These results illustrate how additional entangling operations can increase exposure to gate errors.

5. Custom Processor D Architecture

5.1 Design

Processor D is a custom four-qubit architecture designed for the C1 problem.

Its undirected connectivity edges are:

• (0,1)

• (1,2)

• (2,3)

• (3,0)

• (0,2)

This creates a diamond-like topology with a maximum vertex degree of three.

The supported basis gates are:

`rz`, `sx`, `x`, and `cx`.

5.2 Noise model

The configured error probabilities are:

Error type	Probability
Single-qubit gate	0.001
Two-qubit gate	0.005
Readout	0.01

5.3 Design rationale and validation

The architecture supports a linear entangling chain while providing alternative routes through its additional connections.

The design aims to balance connectivity with the number of physical couplers. Compared with a fully connected four-qubit graph, it uses fewer connections while retaining useful routing alternatives.

The supplied Processor D validation script accepted the configuration for the required four-qubit C1 workload.

Structural validation confirms compliance with the configured constraints. It does not independently prove that Processor D outperforms other architectures.

6. Shot-Based VQE Energy Estimation

6.1 Experimental method

The best converged depth-1 COBYLA parameters were saved and reused to construct the optimized VQE circuit.

The same circuit was evaluated under four measurement conditions:

1. Ideal simulation, 1024 shots per group.
2. Ideal simulation, 8192 shots per group.
3. Noisy Processor D, 1024 shots per group.
4. Noisy Processor D, 8192 shots per group.

The parity-reduced Hamiltonian contains two QWC measurement groups. Therefore, the total shot counts are 2048 and 16384, respectively.

The energy is estimated using the supplied `energy_shots()` function, which also calculates the standard error.

6.2 Results

Condition	Shots/group	Total shots	Energy (Ha)	Standard error (Ha)	Absolute error (mHa)
Ideal	1024	2048	–1.14591123	0.00779005	8.6052
Ideal	8192	16384	–1.13916230	0.00284089	1.8563
Noisy D	1024	2048	–1.11803072	0.00905620	19.2753
Noisy D	8192	16384	–1.11419528	0.00318715	23.1108

6.3 Discussion

The ideal simulation's standard error decreases as the number of shots increases. This is consistent with the reduction in statistical uncertainty expected from additional measurements.

The ideal 8192-shot result has an absolute error of 1.8563 mHa. It is close to, but does not meet, the strict 1.6 mHa chemical-accuracy threshold.

Neither noisy Processor D result meets chemical accuracy.

The noisy estimates remain substantially above the reference energy. Increasing the shot count reduces statistical uncertainty but does not eliminate systematic bias introduced by the noise model.

This distinction is important: additional measurements can improve statistical precision without necessarily improving accuracy when systematic errors dominate.

7. Processor A, B, and D Resource Comparison

7.1 Benchmark methodology

The reduced and unreduced C1 circuits were transpiled for the official Processor A and B topologies and the custom Processor D topology.

The comparison records:

• Logical and physical qubit counts.

• Logical and transpiled CX counts.

• Circuit depth.

• Two-qubit depth.

• SWAP count.

• CX overhead.

The benchmark uses the official Phase 2 Processor A/B definitions from the supplied starter implementation and the validated Processor D configuration.

7.2 Results

Circuit	Processor	Logical qubits	Physical qubits	CX	Depth	Two-qubit depth	SWAPs
JW-4q	A	4	5	3	21	3	0
JW-4q	B	4	9	3	21	3	0
JW-4q	D	4	4	3	21	3	0
Parity-2q	A	2	5	1	13	1	0
Parity-2q	B	2	9	1	13	1	0
Parity-2q	D	2	4	1	13	1	0

7.3 Discussion

For the tested circuits, all three processor architectures produced identical transpiled CX counts and depths. No explicit SWAP gates were required.

The parity-reduced circuit uses fewer logical qubits and CX gates than the Jordan–Wigner circuit.

These results show that qubit reduction can provide a direct circuit-resource advantage for this workload.

However, the absence of routing overhead in these small circuits does not establish that the processors will perform identically for larger or more connectivity-intensive circuits. The results should be interpreted within the scope of the tested C1 circuits.

8. Limitations and Future Improvements

8.1 Limitations

• Only one molecule and one bond length were studied.

• The noise model uses simplified gate and readout error probabilities.

• Simulated noise does not fully represent the behaviour of a real quantum processor.

• The shot-based energy results contain statistical uncertainty and systematic noise bias.

• The architecture comparison covers only the selected benchmark circuits.

• The VQE sweep uses a limited number of ansatz depths, optimizers, and random seeds.

8.2 Future improvements

Potential improvements include:

• Testing additional ansatz structures and optimization algorithms.

• Increasing the number of random seeds for more robust optimizer comparisons.

• Applying readout-error mitigation.

• Investigating zero-noise extrapolation.

• Testing additional molecular geometries.

• Evaluating larger circuits to investigate connectivity and routing overhead more thoroughly.

• Studying alternative Processor D topologies using the same benchmark and noise assumptions.

These are proposed future directions and were not claimed as completed experiments.

9. Conclusion

The Phase 2 C1 experiments demonstrate that parity mapping with two-qubit reduction can preserve the exact H■ ground-state energy while reducing the problem from four qubits to two.

The VQE sweep shows that COBYLA achieves chemical accuracy across all tested configurations with fewer evaluations than SPSA. The depth-1 ansatz provides a more compact circuit for this problem.

Bell-state experiments demonstrate high ideal fidelity and reduced noisy fidelity, particularly for the pair requiring more entangling gates.

The shot-based VQE experiment shows that increasing the measurement count reduces statistical uncertainty, but the noisy Processor D results remain affected by substantial energy bias.

Finally, the A/B/D resource comparison shows that all tested circuits execute without explicit SWAP overhead under the configured transpilation settings.

Overall, the results highlight the importance of qubit reduction, ansatz selection, measurement strategy, and processor connectivity when designing quantum chemistry workflows.

Reproducibility files: `main.ipynb`, `processors/processor_D.json`, `starter/qff_chem.py`, experiment scripts, and the CSV files in `results/tables/`.